How can a network learn useful representations without labels?

$$\mathbf{x} \to g \to \mathbf{h} \to \text{reconstruct} \to \text{fill in} \to \text{contrast} \to \text{measure}$$

Roadmap

1. What is a representation? — and how to measure one
2. Autoencoders — learning by reconstructing the input
3. Masked prediction — learning by filling in what is hidden
4. Contrastive learning — learning which inputs belong together
5. Evaluating representations — k-NN, linear probe, fine-tuning

Practice A after Part 2; Practice B after Part 5.

From architecture to representation

Session 03: a backbone g pretrained on ImageNet transfers to new tasks.

$$\mathbf{x} \to g(\mathbf{x}) = \mathbf{h} \to \text{new head} \to \hat{y}$$

But that backbone needed **1.2 million labeled images**.

What did g actually learn, and could it learn it **without** the labels?

The question for today

> **How can a network learn useful representations without explicit labels?**

Labels are scarce and expensive: 300 labeled X-rays (Session 03 exit question).

Unlabeled images are almost free.

Can the data **supervise itself**?

Part 1 · What is a representation?

1. **What is a representation?** — and how to measure one
2. Autoencoders — learning by reconstructing the input
3. Masked prediction — learning by filling in what is hidden
4. Contrastive learning — learning which inputs belong together
5. Evaluating representations — k-NN, linear probe, fine-tuning

> Session 03: the architecture is an assumption about the input. Today: the objective is an assumption too.

A representation is a change of coordinates

$$\mathbf{x} \in \mathbb{R}^D \longmapsto \mathbf{h} = g(\mathbf{x}) \in \mathbb{R}^k$$

Session 01: the hidden layer bent the input space until **one line** separated the classes.

A good $\mathbf{h}$ makes the task **easy for a simple model**.

What makes a representation useful

- keeps the information the task needs;
- discards nuisances: lighting, position, background;
- is compact;
- is **linearly decodable**: a linear layer can read the answer.

"Useful" always means **useful for some task**. There is no best $\mathbf{h}$ in the abstract.

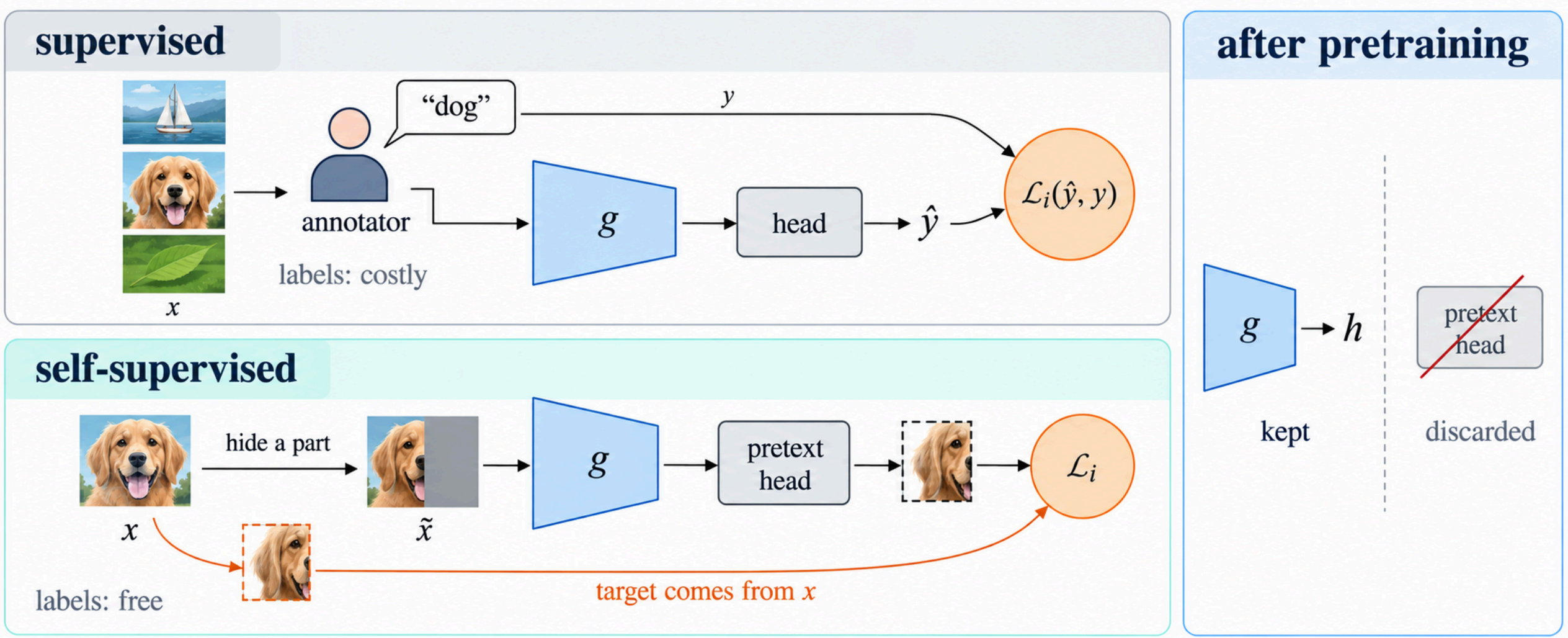
Embeddings

One vector per item; **distance means similarity**.

- Words, users, products: `nn.Embedding` is a lookup table, one-hot $\times$ matrix.
- Images: the output of g. For ResNet-18, $\mathbf{h} \in \mathbb{R}^{512}$.

Two images are "similar" if their embeddings are close:

$$\cos(\mathbf{h}_i, \mathbf{h}_j) = \frac{\mathbf{h}_i^\top \mathbf{h}_j}{\|\mathbf{h}_i\| \, \|\mathbf{h}_j\|}$$


supervised
"dog"
y
annotator
labels: costly
g
head
$\hat{y}$
$\mathcal{L}_i(\hat{y}, y)$
x
self-supervised
hide a part
g
pretext head
$\mathcal{L}_i$
x
$\tilde{x}$
labels: free
target comes from x
after pretraining
g
h
pretext head
kept
discarded

Labels from the data itself

A **pretext task**: hide or alter part of $\mathbf{x}$, predict it from the rest.

Two families:

- **self-prediction** — reconstruct, denoise, fill in;
- **contrastive** — decide which inputs belong together.

The pretext task is **discarded**. The encoder g is **kept**.

Measuring a representation

Freeze g, then read $\mathbf{h}$ with as little as possible:

1. **linear probe** — one linear layer trained on $\mathbf{h}$ (Session 03);
2. **k-NN** — the label of the nearest embeddings, no training at all;
3. **label efficiency** — accuracy as a function of the number of labels.

Probe accuracy measures **representation quality**, not the encoder's own objective.

Prediction before experiment

A ResNet-18 with **random** weights, frozen.

A linear probe on its $\mathbf{h} \in \mathbb{R}^{512}$, trained on CIFAR-10.

Chance is 10 %.

> What accuracy do you expect? Answer in notebook 4.5.

Architecture versus objective

Session	Chooses	Effect on $\mathbf{h}$
03	the architecture of g	what g **can** represent
04	the objective J	what g is **pushed** to represent

Same ResNet-18, different J $\rightarrow$ different $\mathbf{h}$.

Part 2 · Autoencoders

1. What is a representation? — and how to measure one
2. **Autoencoders** — learning by reconstructing the input
3. Masked prediction — learning by filling in what is hidden
4. Contrastive learning — learning which inputs belong together
5. Evaluating representations — k-NN, linear probe, fine-tuning

> Part 1: a representation is judged by what a simple model can do with it.

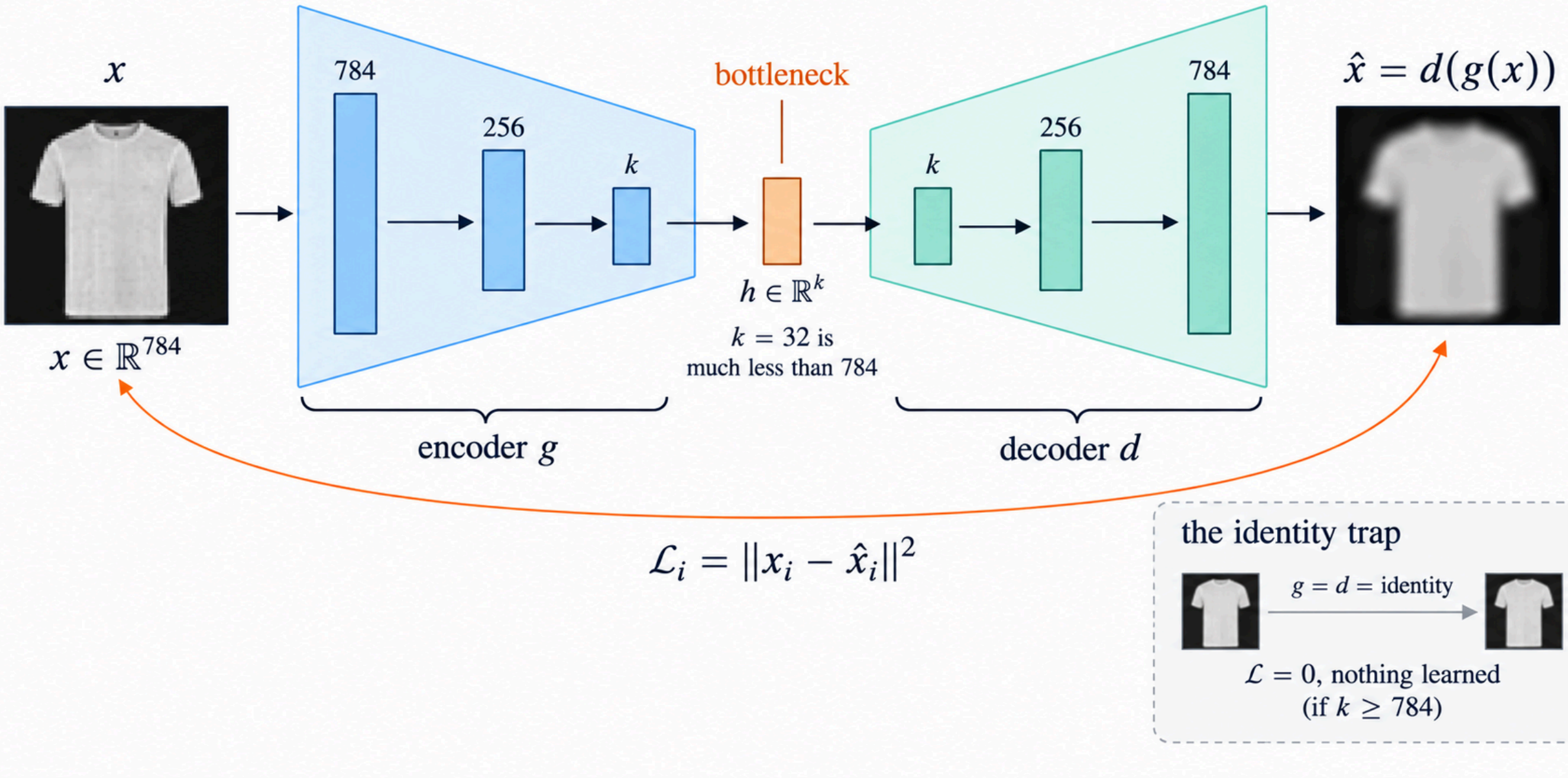
Encoder and decoder

$$\hat{\mathbf{x}} = d\big(g(\mathbf{x})\big), \qquad L_i = \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2$$

The only supervision is the input itself.

The trap: $g = d = \text{identity}$ gives $L = 0$ and learns nothing.

Something must prevent copying.


x
784
256
k
bottleneck
$h \in \mathbb{R}^k$
$k = 32$ is
much less than 784
k
256
784
$\hat{x} = d(g(x))$
$x \in \mathbb{R}^{784}$
encoder g
decoder d
$\mathcal{L}_i = ||x_i - \hat{x}_i||^2$
the identity trap
$g = d =$ identity
$\mathcal{L} = 0$, nothing learned
(if $k \geq 784$)

The bottleneck forces a choice

Undercomplete: $k \ll D$, for example $784 \to 32$.

g cannot pass everything through $\mathbf{h}$, so it must decide **what to keep**.

Compression is the supervision.

A linear autoencoder is PCA

Linear g and d, trained with MSE, learn the subspace of the **top-k principal components**.

Notebook 4.1, Fashion-MNIST, $k = 32$: same test error as PCA (0.015), 31 of 32 directions match.

The only new ingredient of a deep autoencoder is the nonlinearity φ.

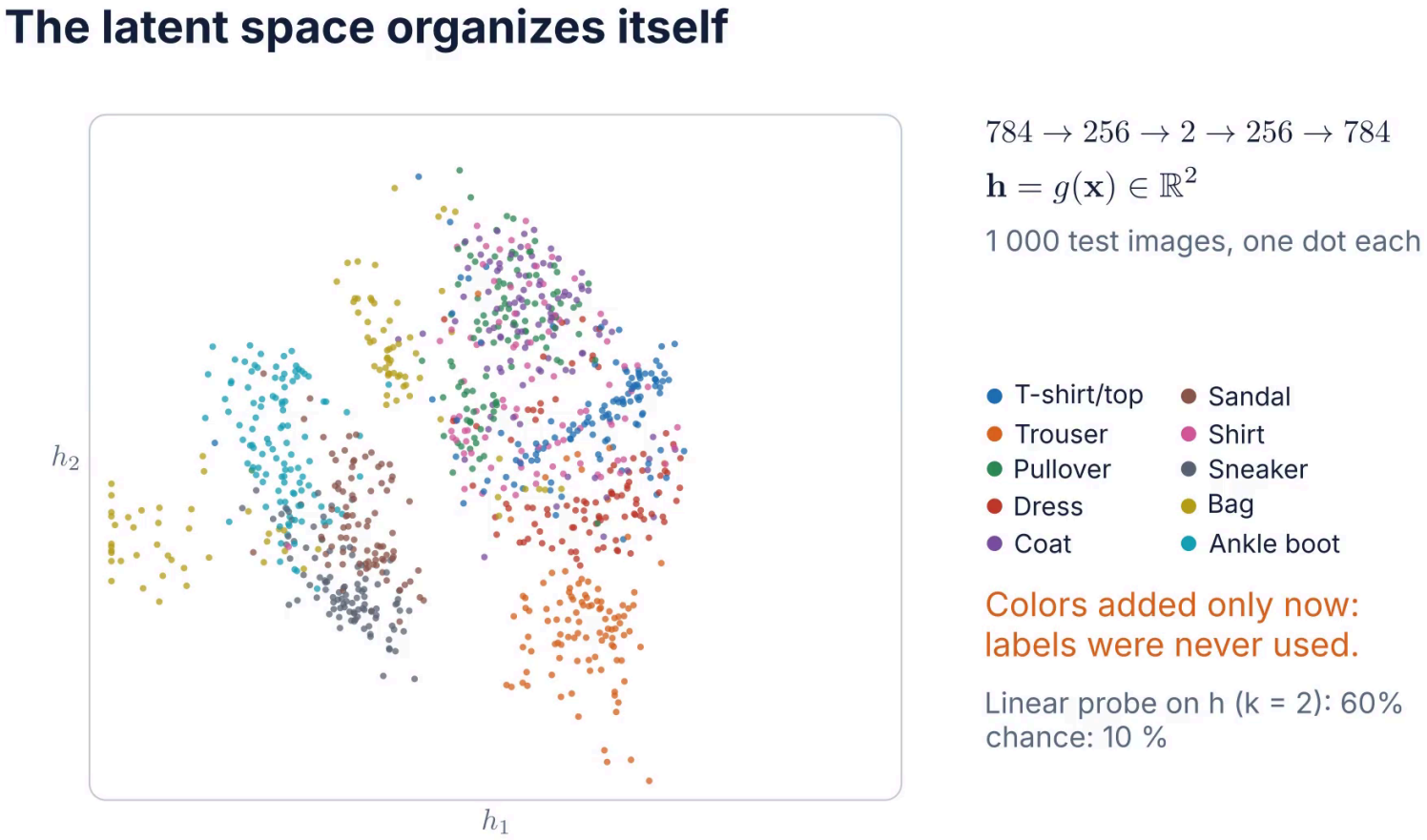
Prediction before experiment

Fashion-MNIST autoencoder, bottleneck $k = 2$ versus $k = 32$.

Which one **reconstructs** better?

In which latent space are the classes **more separated**, judged by a linear probe?

The latent space organizes itself



Open animation

Reconstruction is not the goal

Pixel MSE spends capacity on texture and gives blurry outputs.

Notebook 4.1, $k = 32$:

	MSE	probe
PCA	0.0151	0.807
autoencoder (ReLU)	0.0108	0.812

28 % less reconstruction error buys **0.6 points** of probe.

Convolutional autoencoder

Encoder: the convolutions of Session 03. Decoder: `ConvTranspose2d`, the way back.

$$1 \times 28 \times 28 \to 32 \times 14 \times 14 \to 64 \times 7 \times 7 \to \mathbf{h} \in \mathbb{R}^{32} \to \cdots \to 1 \times 28 \times 28$$

Notebook 4.2: 40 % fewer parameters than the MLP, better reconstruction, **the same probe**.

Denoising autoencoder

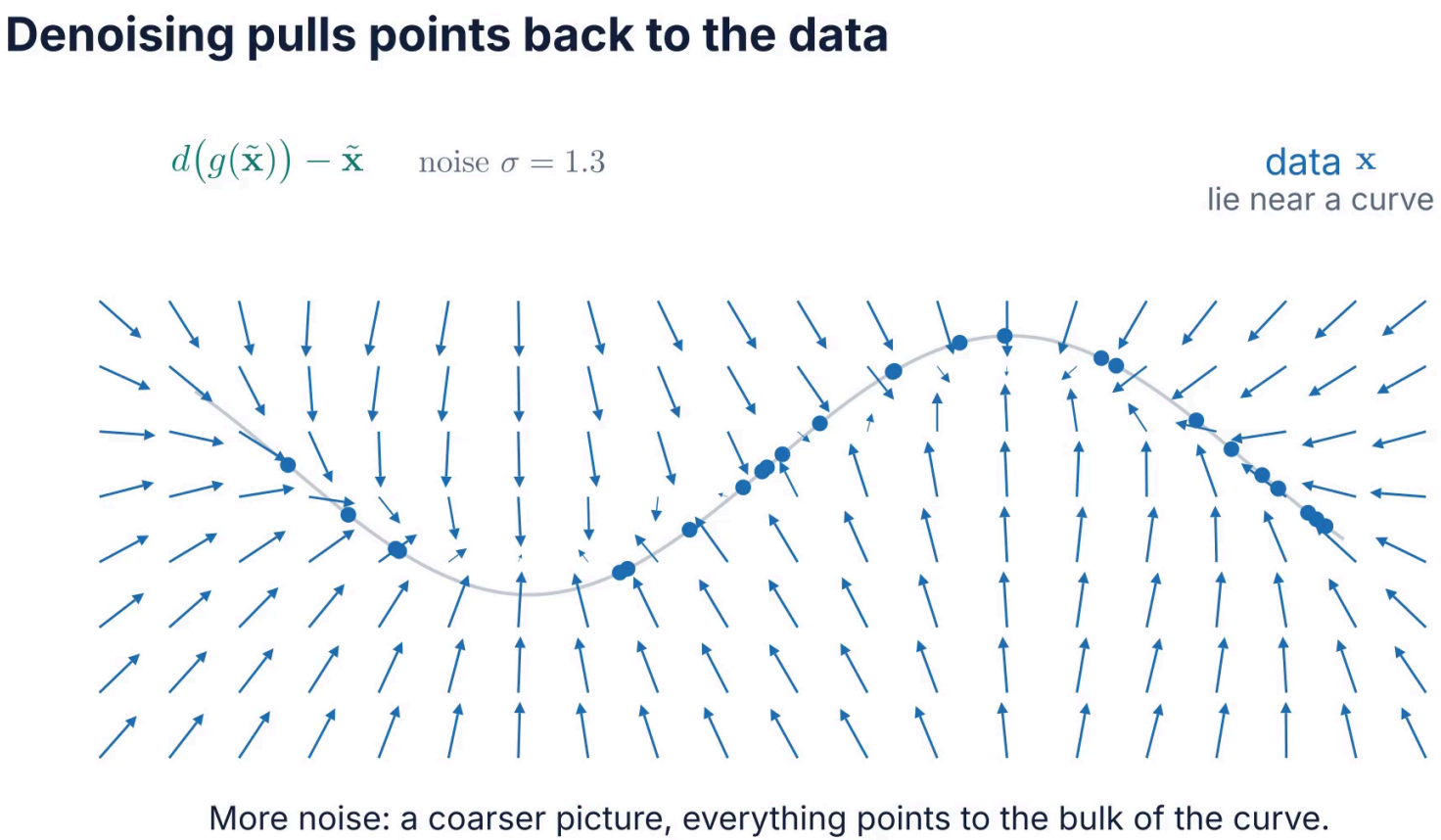
$$\tilde{\mathbf{x}} = \mathbf{x} + \boldsymbol{\varepsilon}, \qquad L_i = \left\| \mathbf{x}_i - d\big(g(\tilde{\mathbf{x}}_i)\big) \right\|^2$$

The encoder sees the corrupted input; the target is the clean one. Copying is no longer possible.

Notebook 4.2, $k = 32$: clean reconstruction **worse** (0.0114 → 0.0126), probe **better** (0.813 → 0.819).

With noise, even a **wide** code works: $k = 1024$ copies without noise (probe = pixels, 0.846) and learns with a mask (0.871).

Denoising pulls points back to the data



Open animation

Other ways to prevent copying

- a narrow bottleneck (undercomplete);
- noise or masks on the input (denoising);
- sparse codes: most units of $\mathbf{h}$ at zero;
- the generative version, the **VAE**: $\mathbf{h}$ is a distribution (beyond this course).

Common idea: **limit the information** that can pass through $\mathbf{h}$.

Practice A — Autoencoders

Worked notebooks on Fashion-MNIST, one change per cell, CPU:

- 4.1 · Autoencoders: $k = 2$ latent space, $k = 2 \ldots 128$, linear AE vs PCA, the identity trap
- 4.2 · Denoising and convolutional: noise and masks, $k = 1024$, learned filters, conv AE

> Write down two numbers per model: reconstruction error and probe. Do they move together?

Part 3 · Masked prediction

1. What is a representation? — and how to measure one
2. Autoencoders — learning by reconstructing the input
3. **Masked prediction** — learning by filling in what is hidden
4. Contrastive learning — learning which inputs belong together
5. Evaluating representations — k-NN, linear probe, fine-tuning

> Part 2: denoising predicts the clean input from a corrupted one. Masking is the extreme case of corruption.

From noise to masks

Instead of perturbing every pixel, **remove** whole regions.

Predict only what was hidden.

$$\mathbf{x} = (\mathbf{x}_{\text{visible}}, \mathbf{x}_{\text{masked}}), \qquad \text{predict } \mathbf{x}_{\text{masked}} \text{ from } \mathbf{x}_{\text{visible}}$$

Hand-designed pretext tasks

Before masking became the standard:

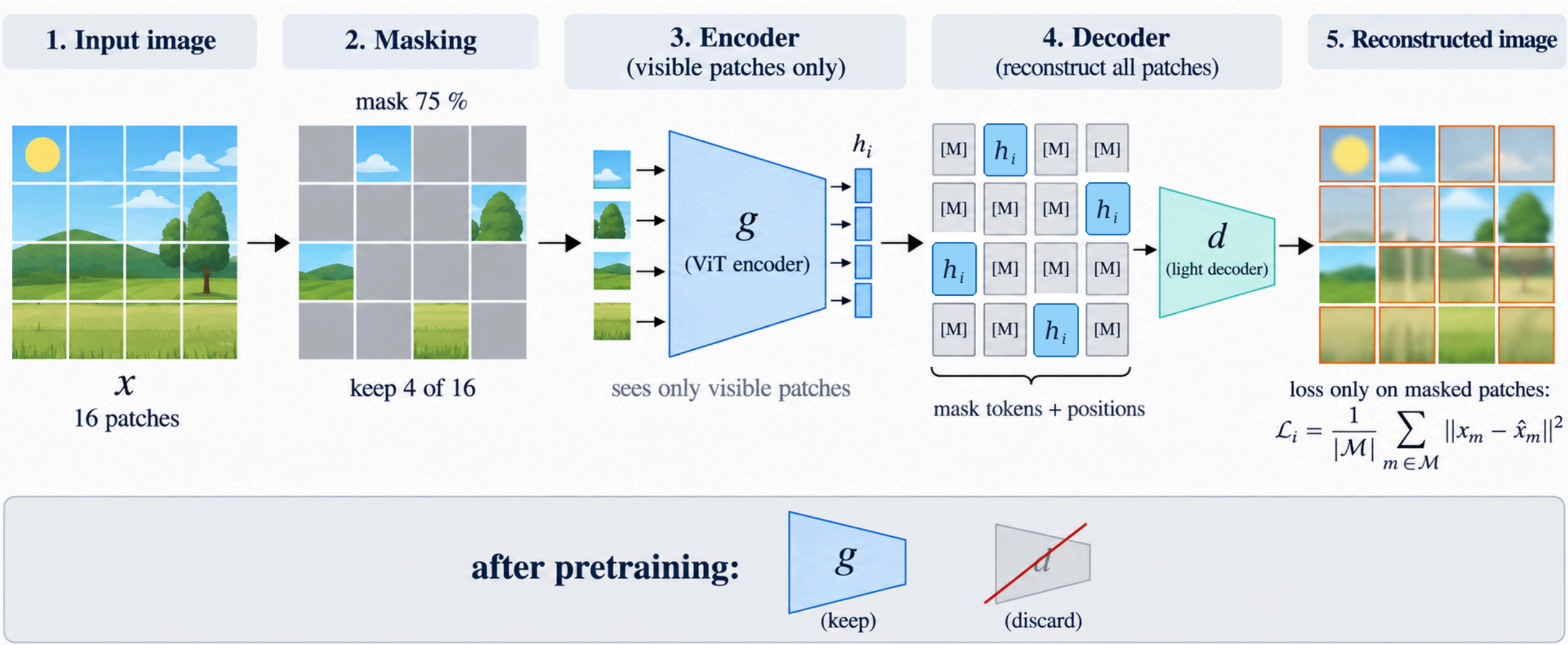
- predict the **rotation** applied to an image (0°, 90°, 180°, 270°);
- solve a **jigsaw** of shuffled tiles;
- **colorize** a grayscale image;
- **inpaint** a missing square.

Each worked a little, and each had **shortcuts**: low-level cues that solve the task without understanding the image.

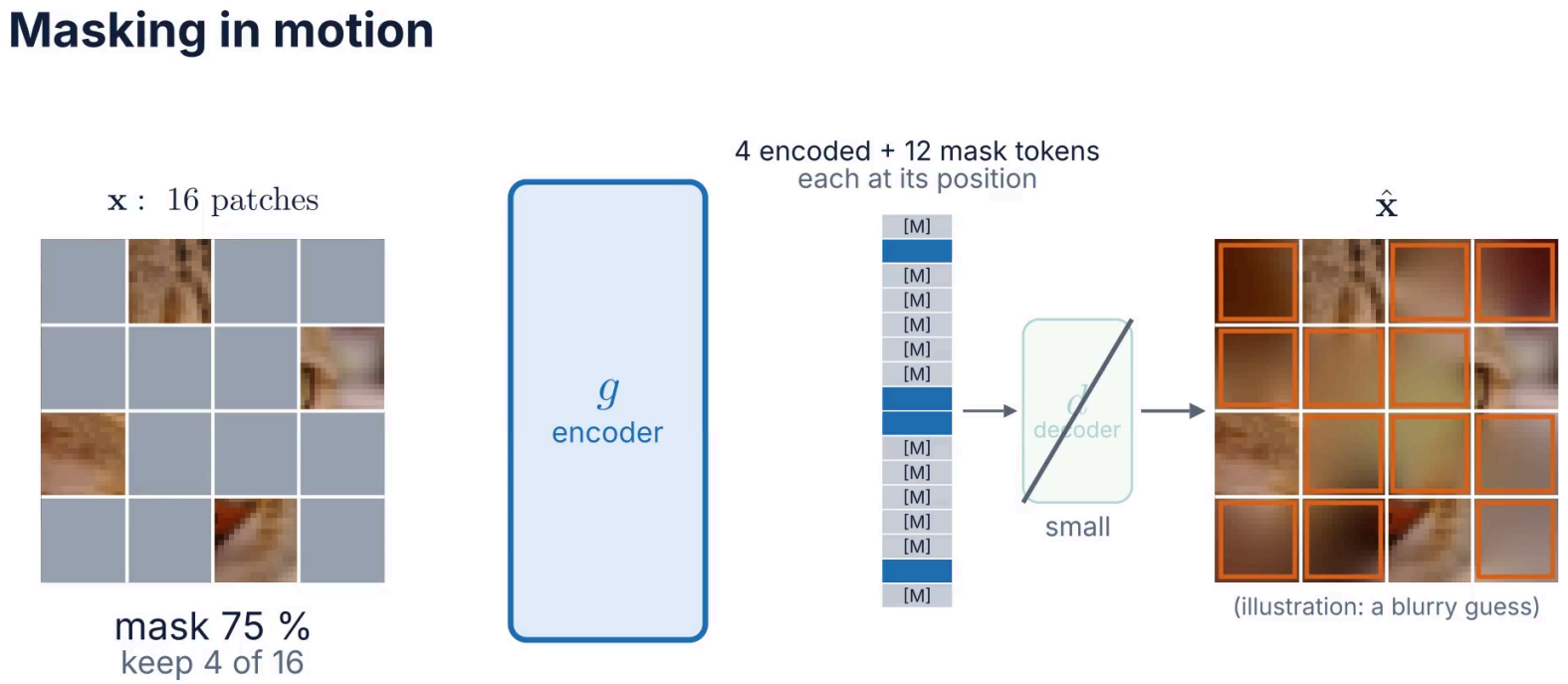
Masked autoencoders

1. split the image into patches (Session 03);
2. mask **75 %** of them at random;
3. the encoder g sees **only the visible patches**;
4. a small decoder d receives mask tokens [M] at the hidden positions and fills them in;
5. the loss counts **only the masked patches**:

$$L_i = \frac{1}{|M|} \sum_{m \in M} \left\| \mathbf{x}_m - \hat{\mathbf{x}}_m \right\|^2$$


1. Input image
2. Masking
mask 75 %
3. Encoder
(visible patches only)
4. Decoder
(reconstruct all patches)
5. Reconstructed image
h_i
g
(ViT encoder)
[M]
h_i
d
(light decoder)
x
16 patches
keep 4 of 16
sees only visible patches
mask tokens + positions
loss only on masked patches:
$\mathcal{L}_i = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} ||x_m - \hat{x}_m||^2$
after pretraining:
g
(keep)
d
(discard)

Masking in motion



After pretraining: keep the encoder g, discard the decoder d.

Open animation

Why mask so much?

Images are **redundant**: with 25 % masked, a hidden patch can be copied from its neighbors.

Text is not: BERT masks only **15 %** of the words (Session 05).

Bonus: the encoder processes only 25 % of the patches, so pretraining is **several times faster** (about 3× in the MAE paper; 1.7× in notebook 4.3, where the decoder is not much smaller).

How much to mask depends on **how redundant the data are**.

Prediction before experiment

Tiny MAE on Fashion-MNIST: 16 patches of 7×7, Transformer encoder.

Mask **25 %** or **75 %**.

Which one gives the better **linear probe**?

Which one **trains faster**?

What masked prediction learns

The information is in $\mathbf{h}$, but **not linearly arranged**:

- MAE paper, ViT-L on ImageNet: the linear probe stays about **10 points** below fine-tuning;
- notebook 4.3, 100 labels per class: probe 0.77, fine-tuning 0.78; pretraining adds **10 points** to the probe over a random encoder.

Masked features shine **after fine-tuning**. Next: an objective whose features read well **linearly**.

Part 4 · Contrastive learning

1. What is a representation? — and how to measure one
2. Autoencoders — learning by reconstructing the input
3. Masked prediction — learning by filling in what is hidden
4. **Contrastive learning** — learning which inputs belong together
5. Evaluating representations — k-NN, linear probe, fine-tuning

> Part 3: self-prediction spends effort on pixels. Alternative: no decoder at all.

Learning relations, not pixels

In embedding space:

- **similar** inputs → close;
- **dissimilar** inputs → far.

No reconstruction, no decoder.

The question is no longer "what is in this image?" but "**is this the same image?**"

Augmentations define "the same"

Two random views of one image $\mathbf{x}$:

$$\tilde{\mathbf{x}}_i = t(\mathbf{x}), \quad \tilde{\mathbf{x}}_j = t'(\mathbf{x}), \qquad t, t' \sim \mathcal{T}$$

are a **positive pair**.

The augmentations chosen are the **invariances learned**: whatever $\mathcal{T}$ changes, $\mathbf{h}$ learns to ignore.

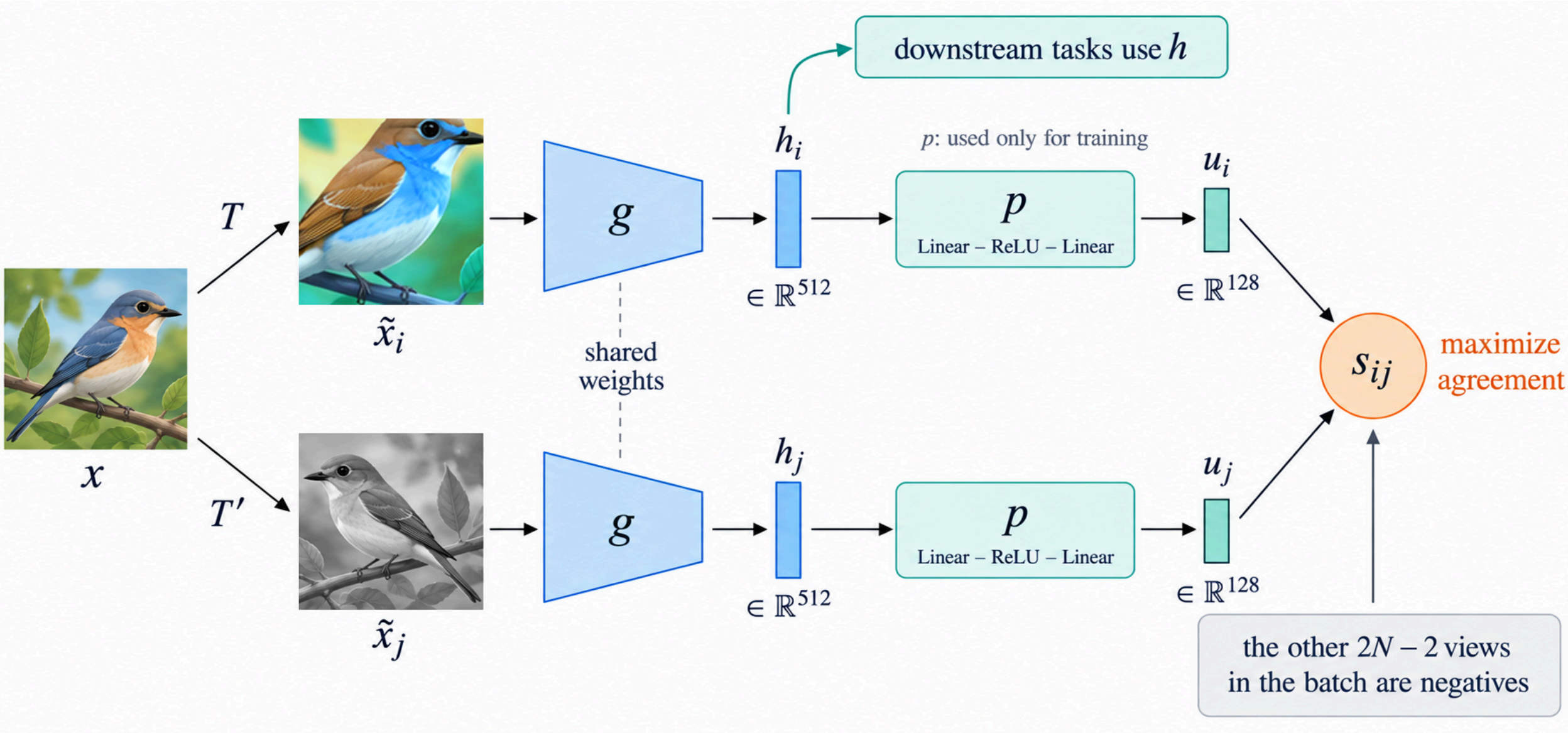
Session 03: augmentation encodes invariances. Here it is the **only** supervision.

SimCLR

For a batch of N images:

1. two views of each image: $2N$ views;
2. shared encoder: $\mathbf{h} = g(\tilde{\mathbf{x}})$;
3. projection head: $\mathbf{u} = p(\mathbf{h})$;
4. each view must find **its** other view among the other $2N - 1$.

The other $2N - 2$ views are the **negatives**.


downstream tasks use h
h_i
p: used only for training
u_i
T
g
p
Linear – ReLU – Linear
$\in \mathbb{R}^{512}$
$\in \mathbb{R}^{128}$
$\tilde{x}_i$
shared weights
s_{ij}
maximize agreement
x
T'
h_j
u_j
g
p
Linear – ReLU – Linear
$\in \mathbb{R}^{512}$
$\in \mathbb{R}^{128}$
$\tilde{x}_j$
the other $2N-2$ views in the batch are negatives

Similarity and temperature

$$s_{ij} = \frac{\mathbf{u}_i^\top \mathbf{u}_j}{\tau \left\|\mathbf{u}_i\right\| \left\|\mathbf{u}_j\right\|}$$

Cosine similarity, divided by a **temperature** τ.

Small τ sharpens the comparison and focuses on the **hardest negatives**: the ones that are most similar.

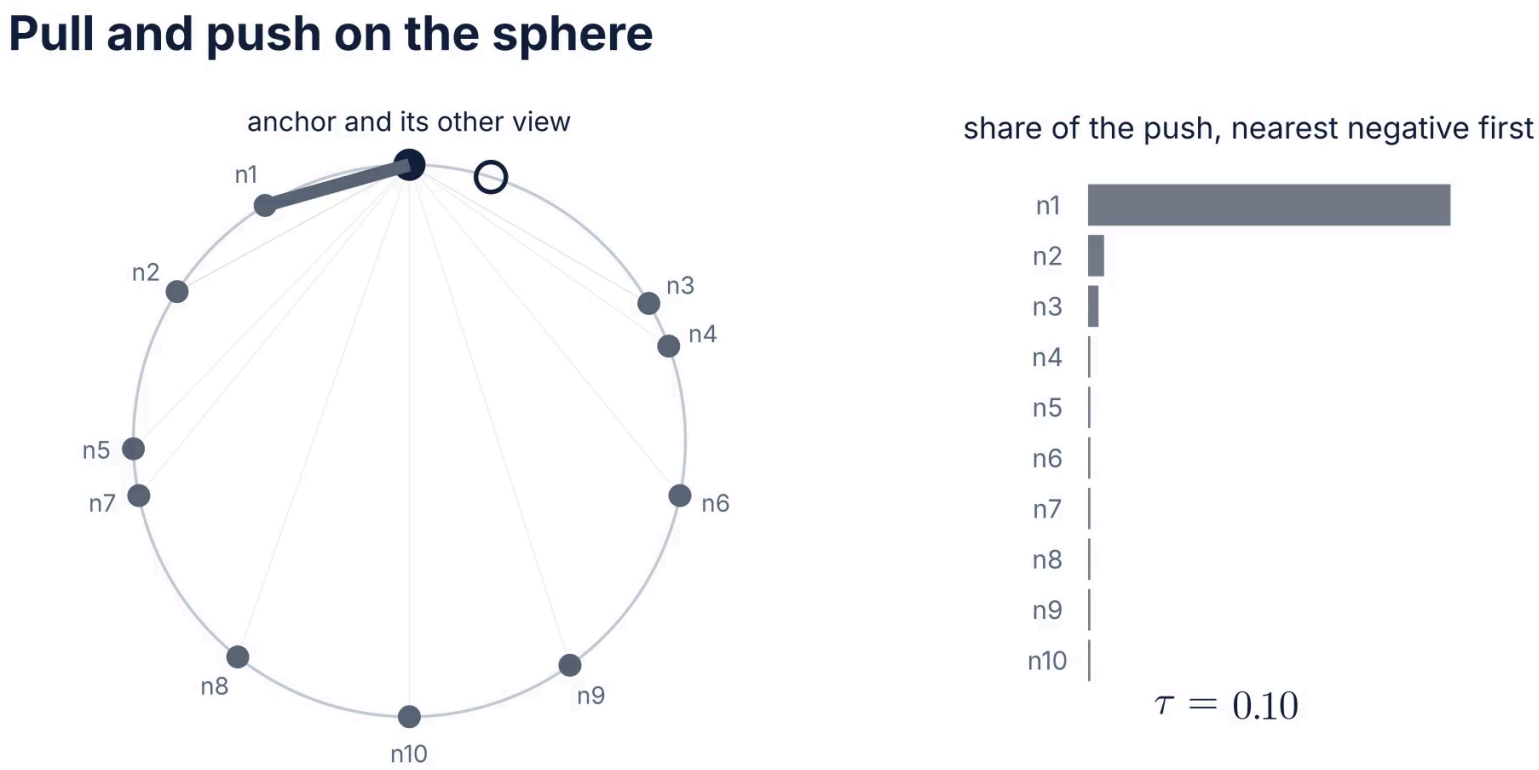
InfoNCE is a cross-entropy

$$L_i = -\log \frac{\exp\left(s_{i,j(i)}\right)}{\sum_{k \neq i} \exp(s_{ik})}$$

A classification over $2N - 1$ candidates, where the correct "class" is the other view $j(i)$.

The same softmax + cross-entropy as Session 01. At initialization, $L \approx \ln(2N - 1)$.

Pull and push on the sphere



Here the embeddings move directly; in SimCLR the gradient flows into g and p.

Small τ: the nearest negatives do almost all the pushing. Large τ: every negative pushes a little.

Open animation

Prediction before experiment

Train SimCLR **without color jitter**.

The pretext task: easier or harder? Does the loss end lower or higher?

The representation: does the k-NN accuracy go up or down?

Second question: $\tau = 0.05$ versus $\tau = 1.0$.

Why a projection head?

The loss makes $\mathbf{u}$ **invariant** to the augmentations, so $\mathbf{u}$ throws information away: color, orientation, position.

$\mathbf{h}$, one layer earlier, keeps more of it.

Downstream we use $\mathbf{h}$ and **discard** p.

Negatives and batch size

More negatives → a harder task → better features.

SimCLR needs **large batches**: $N = 512$ gives 1 023 candidates per view.

MoCo keeps a **queue** of past embeddings instead, so the negatives do not have to fit in one batch.

Collapse

If every $\mathbf{u}$ is the same, the two views of an image always match.

Negatives prevent this: a constant $\mathbf{u}$ cannot be far from the other images.

Methods **without** negatives (BYOL, DINO) prevent collapse with asymmetry: a second network updated slowly, and no gradient through one branch.

Symptom to watch: the variance of $\mathbf{h}$ across a batch goes to zero.

The same idea, other pairs

Method	Positive pair	Against collapse
SimCLR	two views of one image	negatives in the batch
MoCo	two views of one image	a queue of negatives
BYOL, DINO	two views of one image	slow teacher network
CLIP	an image and its **caption**	negatives in the batch

CLIP is a **foundation model**: Session 05.

Part 5 · Evaluating representations

1. What is a representation? — and how to measure one
2. Autoencoders — learning by reconstructing the input
3. Masked prediction — learning by filling in what is hidden
4. Contrastive learning — learning which inputs belong together
5. **Evaluating representations** — k-NN, linear probe, fine-tuning

> Part 4: three objectives — reconstruct, fill in, contrast — and one encoder g.

Three ways to read a representation

Reading	What is trained	What it measures
k-NN	nothing	neighborhoods in $\mathbf{h}$
linear probe	one linear layer	linear decodability
fine-tuning	everything	g as a starting point

The same three strategies as Session 03, now applied to **our own g**.

k-NN in embedding space

For a new image: find the k training images with the most similar $\mathbf{h}$ (cosine), and vote their labels.

No training, no hyperparameters beyond k.

It is also **retrieval**: "show me images like this one".

Label efficiency

The question that justifies self-supervision:

> How much accuracy per label?

Notebook 4.5: linear probe with 1, 10, 100, 1000 and 5000 labels per class, for four ResNet-18 encoders — random, autoencoder, SimCLR, ImageNet.

The gap at the **left** of the curve is the value of the unlabeled images.

Looking at embeddings

t-SNE and UMAP keep **neighbors**, not distances.

- cluster sizes mean nothing;
- the gaps between clusters mean nothing;
- another seed gives another picture.

Use them to **look**. Use the probe and k-NN to **measure**.

Prediction before experiment

You have the SimCLR encoder and **10 labels per class**: 100 images in total.

Which wins on the test set?

- k-NN on $\mathbf{h}$;
- linear probe on $\mathbf{h}$;
- fine-tuning all of g with a new head.

> And why? Answer in notebook 4.5, section 6.

Practice B — Contrastive learning and evaluation

Worked notebooks on CIFAR-10, GPU:

- 4.4 · SimCLR: two views, InfoNCE checked against the formula, 3 epochs live, the 100-epoch checkpoint, the color shortcut
- 4.5 · Evaluating representations: k-NN, retrieval, probe, label efficiency, t-SNE, 10 labels per class

At home: 4.3 · Masked prediction, a tiny MAE on Fashion-MNIST.

> On Kaggle, add the dataset `si7011-s04-simclr` before running 4.4.

Integrating exercise — EuroSAT without labels

4.6 · Integrating exercise · the satellite images of Session 03, the same split and contract

- design the two views for satellite images, write InfoNCE;
- pretrain SimCLR on the images **without their labels**, logged to **MLflow**;
- evaluate with k-NN and probe using **1 %** and **10 %** of the labels, against ImageNet;
- `embed_raw`: an embedding service for new raw images.

> With few labels: photos **with** labels, or satellite images **without** them?

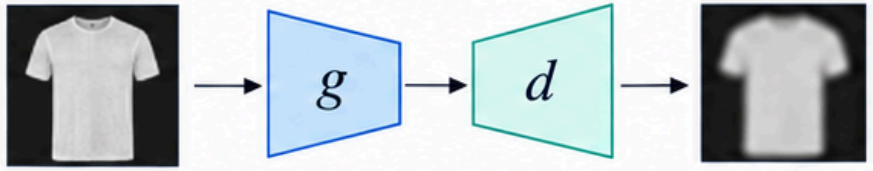
Choosing a pretext task

Objective	Learns	Needs	Reads best with
autoencoder	what is needed to redraw pixels	a bottleneck	—
denoising	the structure of the data	a corruption	probe
masked	how parts relate	a decoder, redundancy	fine-tuning
contrastive	what augmentations do not change	good augmentations, many negatives	k-NN, probe

What each objective assumes

	Assumes	Pays with
reconstruction	every pixel matters	capacity spent on texture
denoising	the noise is not the signal	the noise we chose
masked prediction	parts predict parts	features that need fine-tuning
contrastive	augmentations do not change meaning	the augmentations we chose

reconstruct (autoencoder)

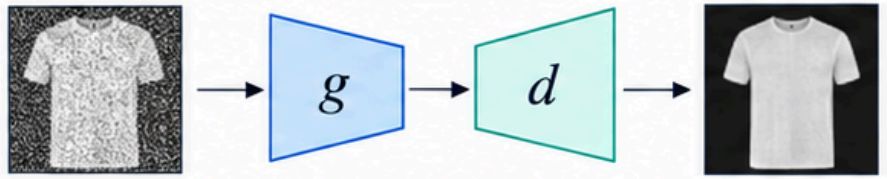


x $\hat{x}$

$$\|x - \hat{x}\|^2$$

keeps: everything needed to redraw the pixels, texture included.

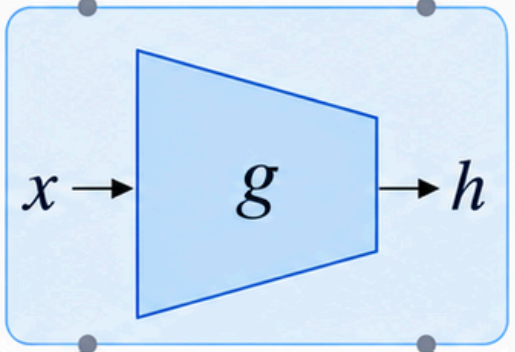
denoise



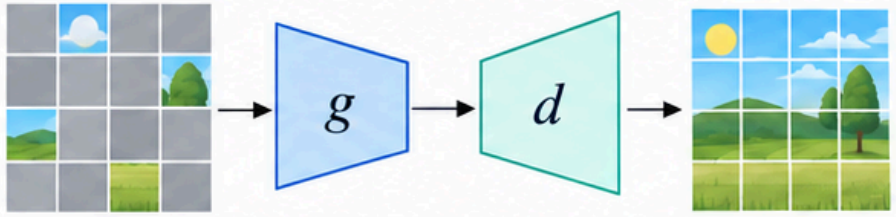
$\tilde{x} = x + \epsilon$ $\hat{x}$

$$\|x - d(g(\tilde{x}))\|^2$$

keeps: the structure of the data, not the noise.



fill in (masked)

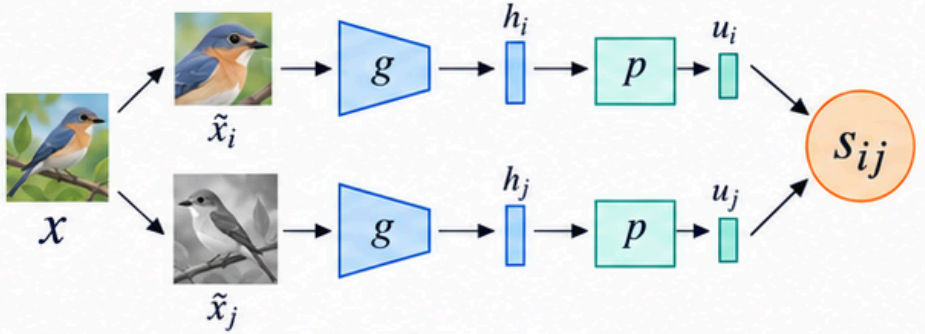


masked x $\tilde{x}$

$$\sum_{m \in \mathcal{M}} \|x_m - \hat{x}_m\|^2$$

keeps: how the parts of an input relate.

contrast



x $\tilde{x}_i$ $\tilde{x}_j$ h_i u_i h_j u_j

InfoNCE over s_{ij}

keeps: what the augmentations do not change.

the pretext task is discarded · g is kept · measure it with k-NN, linear probe, fine-tuning.

Exit question

50 000 unlabeled dermatology images, 200 labeled.

Choose two and justify:

- SimCLR with the default augmentations;
- a masked autoencoder;
- ImageNet backbone + linear probe;
- a denoising autoencoder;
- first check what color jitter does to a skin lesion, then pretrain.